# Sathmi Peiris

**Computer Engineering Graduate** - University of Sri Jayewardenepura

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## RESEARCH INTERESTS

My research interests are in **machine learning, computer vision, edge AI**, and **robotics**, along with **scientific machine learning** applied to science.

## EDUCATION

**Bachelor of the Science of Engineering Honours (BScEngHons)**| **Link** Feb 2022 – July 2026
Major in Computer Engineering; Minor in Data Management
University of Sri Jayewardenepura, Sri Lanka

- **GPA:** 3.88/4.00 (First-Class Honours) | Transcript | **Rank :** 1/170 (Faculty), 1/41 (Dept.)
- **Gold Medals (52nd Convocation of University of Sri Jayewardenepura, Sri Lanka):** Most Venerable Balangoda Ananda Maitreya Gold Medal (Faculty) | [View] & Gold Medal for Computer Engineering (Dept) | [View]
- **Dean's List** (3.8+) for 6/8 semesters.
- The complete degree program was conducted and assessed in English medium.

**Anula Vidyalaya Nugegoda**, Grade 1-13 Jan 2007 – Aug 2020

- G.C.E. Advanced Level Examination: Combined Mathematics, Chemistry, Physics (AAB) | [Statement]

## EXPERIENCE

**Temporary Demonstrator** Oct 2026 – Present
The Open University of Sri Lanka (Dept. of Electrical & Computer Engineering)

- Facilitate practical sessions, instruct, and evaluate engineering students across core modules including Software Development for Engineers (EEK3218), Introduction to Computer and Communication Engineering (EES3236), Data Communications and Networks (EES5334), and Data Structures and Algorithms (EES4235).

**Volunteer Research Assistant** Sept 2026 – Present
Sri Lanka Institute of Information Technology (SLIIT), Malabe

- Conduct computer vision research focused on analyzing the degradation of thin-layer films utilizing Optical Coherence Tomography (OCT) images.

**Trainee Full-Stack Developer & DevOps** | [Service Letter] Apr 2025 – Oct 2025
Sri Lanka Telecom PLC (Digital Platforms)

- Completed a six-month professional placement in the Digital Platforms Development Section, featuring Agile project rotations across DevOps infrastructure and MERN stack development.
- Built responsive UIs and core components for the **Debt Recovery System (DebtX)**, a platform automating legal demand generation for thousands of cases.
- Refactored backend RESTful APIs (Node.js/Express) for performance optimization, managed MongoDB data modeling, and integrated a PDF engine using `pdf-lib` and React.
- Supported CI/CD pipelines via Git/GitHub/Bitbucket, tested and documented APIs using Postman and Swagger, and provisioned a **Kubernetes (K8s) cluster** for containerized deployments to maintain system uptime.

## PUBLICATIONS

[C 1] **A Comparative Study of Tabular, Geometric, and Lorentz-Aware Models for Higgs Event Classification**
MERCon 2025, 12th international conference, University of Moratuwa | Paper | Poster
Sathmi Vihansa Peiris, Yethmi Sehansa Peiris, Lakshan Madhushanka Bandara
*Contribution: Conceptualization, methodology formulation for Lorentz-equivariant transformer architectures, and physics-informed feature engineering using CERN ATLAS Open Data.*

## SELECTED PROJECTS

**ROS-Based Deep Learning Framework for Elderly Care (Edge AI)** | Project 2026

- Engineered a distributed, containerized robotics framework on **ROS 2 Humble** running on an **NVIDIA Jetson AGX Orin** edge device and a **TurtleBot 2.0** mobile base to assist the elderly.
- Developed a dual-stage biometric fusion tracking system utilizing **YOLOv8-pose** for body detection, **FaceNet** for facial confirmation, and **OSNet** for continuous feature embedding.
- Integrated a responsive, offline voice interaction layer using **Whisper** for speech-to-text, a local 4-bit quantized **Gemma-3-4B** language model via `llama.cpp`, alongside **Piper** and **espeak-ng** for text-to-speech synthesis.
- Implemented a privacy-preserving fall detection pipeline using a custom lightweight **FallDetectionNet** model, and integrated a custom 5-DOF robotic arm utilizing **ROS MoveIt** for safe, automated medication dispensing.
- *Contribution: Architected the modular ROS 2 framework and behavioral state machine, engineered the dual-stage person re-identification (ReID) pipeline, and developed the LLM-driven conversational interface and real-time emergency alerting subsystems.*

**Deep Learning for High-Energy Collision Classification (Scientific ML)** | Poster 2026

- Classified rare Higgs Boson decay signals amidst vast Standard Model background noise using **CERN ATLAS 13 TeV Open Data**.
- Architected a novel **Lorentz Equivariant Physics Transformer (LEPT)** and corrected Minkowski four-vector mass assignments in existing Graph Neural Networks.
- *Contribution: Led the mathematical formulation of the Lorentz Equivariant Physics Transformer (LEPT), corrected baseline Minkowski mass assignments, and conducted SHAP-based model interpretability analysis to ensure strict adherence to fundamental physical conservation laws.*

**Smart Gas Distribution Platform (Microservices & Full-Stack)** | Project 2026

- Developed the frontend and backend for a microservices-based LPG distribution platform to mitigate physical queue chaos through virtual token issuance and live stock tracking.
- Orchestrated 6 **Spring Boot** microservices, an API Gateway, and Eureka Registry using **Docker**, **Kafka**, and isolated **PostgreSQL** databases.
- *Contribution: Implemented role-based workflows secured with **JWT** authentication and developed both the inventory microservice and the user interface.*

**Sri Lankan Car Price Prediction & Market Analysis System** | Project 2025

- Developed an end-to-end machine learning system and web dashboard using **Next.js** and **Flask** to predict used car prices via real-time scraped data from Riyasewana and Ikman.
- Engineered 20+ features incorporating economic indicators (USD/LKR rates, fuel prices, inflation) and evaluated 11 ML algorithms on Modal.com cloud utilizing over 10,000 listings.
- Achieved an $R^2$ of 92.35% (96% of predictions within $\pm$20 Lakhs) using a **Random Forest** model with 300 trees, optimizing the architecture down to 12 core features.
- *Contribution: Implemented the real-time scraping pipeline, performed time-series forecasting with 40+ analytical plots, corrected data leakages, and managed the application rollout using PowerShell deployment commands.*

## TECHNICAL STRENGTHS

- **Programming Languages:** JavaScript (MERN Stack, Next.js), Python, C++, Java, SQL, MATLAB
- **Libraries & Frameworks:** PyTorch, TensorFlow, Keras, OpenCV, YOLOv8, FaceNet, OSNet, Scikit-Learn, Faster Whisper, ROS 2 (Nav2, MoveIt), Flask
- **Digital Design & Hardware:** CUDA C, Verilog, Assembly (8051), Keil uVision, NVIDIA Jetson, FPGA Concepts
- **DevOps & Other:** Docker, Kubernetes, Git, Linux/Bash, PowerShell

## COURSES & TRAINING

**System Verilog for ASIC/FPGA Design & Simulation** | *University of Moratuwa* | [View] 2025
**Fundamentals of Digital System Design** | *University of Moratuwa* | [View] 2024

**Ceylon German Technical Training Institute (CGTTI)** 2021 – 2023

- **Auto Electricity (EA)**
- **Auto Electronics (E3)** | [View]
- **Workshop Practice (WP1)** | [View]

**Certificate in Computer Science** | *NIBM* | [View] 2021

## ACHIEVEMENTS

- **Most Venerable Balangoda Ananda Maitreya Gold Medal:** Awarded for obtaining the Highest GPA in the Faculty of Engineering (2026). | [View]
- **Gold Medal for Computer Engineering:** Awarded for obtaining the Highest GPA in the Department of Computer Engineering (2026). | [View]
- **Exito 2026 Inter-University Robotics & Innovation Challenge:** 2nd Place. | [View]
- **IEEE Xtreme 18.0 (2024):** Ranked 68th in Sri Lanka (Algorithmic Problem Solving). | [View]
- **Xcelerate Robotics Competition:** 2nd Place (Line Following Robot Control).

## VOLUNTEERING & LEADERSHIP

**Editorial Team Head**, IEEE USJ Computer Society | [View] 2024

- Managed the editorial team responsible for curating technical content on emerging technologies.

**Organizing Committee Member**, IEEEXtreme 17.0 & 18.0 2023 – 2024

- Coordinated the programming competition for local university teams.

**Media Panel Member**, CENSOC 2023 – 2024

- Created digital content and managed media coverage for computer engineering society events.

## LANGUAGES

- **Sinhala:** Native **English:** Professional **French:** A1 | [View]
- **Chinese:** HSK 2 | [View] **Hindi:** Intermediate | [View]

## REFERENCES

**Dr. Udaya Wijenayake**
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Faculty of Engineering, University of Sri Jayewardenepura, Sri Lanka.
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**Sathmi Peiris**
November 2, 2026